CSC 102– Discrete Structures I

Logic

Mind Exercise

You find yourself on an island where there are two types of inhabitants: Knights and Knaves. Knights always tell the truth. Knaves always lie. You encounter two individuals, a girl and a boy, but you don't know which one is a knight and which one is a knave. The girl said. “He is knight”. The boy said “The two of us are opposite types?” What is the girl and the boy (***Knight or a Knave***)? Explain.

Logic

Study of reasoning

What is the significance of Logic?

It helps us have effective problem-solving not only in mathematics but in life.

What is Propositional Logic?

What is Proposition?

Proposition

A declarative sentence (declares a fact or stating argument) which can be either **true** or **false** but cannot be both.

Is it a proposition?

What time is it ?

Is it a proposition?

$$x + 5 = 2$$

Is it a proposition?

$$1 + 5 = 2$$

Is it a proposition?

Send me an email.

Is it a proposition?

CSU is located in Mindanao.

Is it a proposition?

Eating vegetables is not good for our health.

What is Propositional Logic?

Propositional Logic

a way of thinking and reasoning using propositions. Propositions can be combined using words like **and**, **or,** **not**, **if-then**, and **if and only if** to create more complex statements.

Example of Propositional Logic

It is raining **and** it is cold.

Compound Prepositions

It is raining **and** it is cold.

Compound Prepositions

It is raining **and** it is cold.

P

Q

Compound Prepositions

It is raining **and** it is cold.

P

Q

P **and** Q

Logical Operators

1) Negation
2) Conjunction
3) Disjunction
4) Exclusive OR
5) Implication
6) Biconditional

Negation

The negation of **p** denoted by **¬p**, is the proposition **not p**.

Negation

$\neg p$ = "It is not the case that p"

Example:

p: Manila is the capital of the Philippines.

$\neg$**p**: It is not the case that Manila is the capital of the Philippines.

Negation

¬p = “It is not the case that p”

Example:

p: Manila is the capital of the Philippines.

¬p: It is not the case that Manila is the capital of the Philippines
¬p : Manila is not the capital of the Philippines.

Negation

p	$\neg p$
T	F
F	T

Conjunction

Let p and q be two propositions. Conjunction of **p** and **q** is denoted by **p ^ q**. When both **p** and **q** are true, then only the compound proposition **p ^ q** is true.

Conjunction: Truth Value

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction / Inclusive OR

Let p and q be two propositions. Disjunction of **p** and **q** is denoted by **p V q**. When both **p** and **q** are false, then only the compound proposition **p V q** is false.

Disjunction: Truth Value

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive OR (XOR)

Let p and q be two propositions. Exclusive OR of **p** and **q** is denoted by **p** $\oplus$ **q**. This means exactly one of **p** and **q** will be true, but both cannot be true.

Exclusive OR (XOR) Example

When you study hard in this subject, you will get a rating of 1 or you will get a rating of 1.25.

Exclusive OR: Truth Value

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication

Let p and q be two propositions. The proposition "If **p** then **q**" denoted by **p → q** is called implication or conditional statement.

Implication

"If **p** then **q**"

⇩

Hypothesis/conditional statement

Implication

"If **p** then **q**"

⬇

Conclusion / Consequent

Implication Truth Value

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Implication Example

“ If the Mathematics Department gets an additional $60,000, then it will hire one new faculty member.

Dean’s Statement

What is the hypothesis ?

If the Mathematics Department gets an additional \$60,000,
then it will hire one new faculty member.

What is the hypothesis ?

If the Mathematics Department gets an additional \$60,000, then it will hire one new faculty member.

p: The Mathematics Department gets an additional \$60,000.

What is the conclusion ?

If the Mathematics Department gets an additional \$60,000, then it will hire one new faculty member.

p: The Mathematics Department gets an additional \$60,000.

What is the conclusion ?

If the Mathematics Department gets an additional \$60,000, then it will hire one new faculty member.

p: The Mathematics Department gets an additional \$60,000.

q: The Mathematics Department will hire one new faculty member.

Case 1 : p and q are true

p : The Mathematics Department gets an additional \$60,000,

q : The Mathematics Department will hire one new faculty member.

Case 1 : p and q are true

p: The Mathematics Department gets an additional \$60,000,

q: The Mathematics Department will hire one new faculty member,

If the Mathematics Department gets an additional \$60,000, then it will hire one new faculty member.

Dean's Statement is true

Case 2 : p is true and q is false

p: The Mathematics Department gets an additional \$60,000,

×q: The Mathematics Department will hire one new faculty member,

Case 2 : p is true and q is false

p: The Mathematics Department gets an additional \$60,000,

×q: The Mathematics Department will hire one new faculty member.

If the Mathematics Department gets an additional \$60,000, then it will hire one new faculty member.

Dean's Statement is false

Case 3 : p is false and q is true

× p: The Mathematics Department gets an additional \$60,000,

q: The Mathematics Department will hire one new faculty member.

Case 3 : p is false and q is true

p: The Mathematics Department gets an additional \$60,000,

q: The Mathematics Department will hire one new faculty member.

If the Mathematics Department gets an additional \$60,000, then it will hire one new faculty member.

Dean's statement will still be true

Case 4 : p is false and q is false

- ×p: The Mathematics Department gets an additional \$60,000,
- ×q: The Mathematics Department will hire one new faculty member,

Case 4 : p is false and q is false

× p : The Mathematics Department gets an additional \$60,000,

× q : The Mathematics Department will hire one new faculty member,

If the Mathematics Department gets an additional \$60,000,
then it will hire one new faculty member.

Dean's statement will still be true

True of false ?

Assume : p = true, q = false and r = true

$$\mathbf{p \wedge q \rightarrow r}$$

True of false ?

Assume : p = true, q = false and r = true

$$p \lor q \rightarrow \neg r$$

True of false ?

Assume : p = true, q = false and r = true

$$p \wedge (q \rightarrow r)$$

True of false ?

Assume : p = true, q = false and r = true

$$p \rightarrow (q \rightarrow r)$$

Implication Representations

“If p then q”

Implication Representations

“p implies q”

Implication Representations

“q when p”

Implication Representations

“q whenever p”

Implication Representations

“q follows from p”

Implication Representations

“p only if q”

Implication Representations

“q is necessary for p”

Implication Representations

“p is sufficient for q”

Implication Representations

“q unless $\neg$ p”

Restate to Conditional Proposition

Mary will be a good student if she studies hard.

Restate to Conditional Proposition

Mary will be a good student if she studies hard.

If Mary studies hard, then she will be a good student.

Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

If John takes calculus, then he has sophomore, junior, or senior standing.

Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

If John has sophomore, junior, or senior standing, then he takes calculus,

Restate to Conditional Proposition

John takes calculus only if he has sophomore, junior, or senior standing.

If John takes calculus, then he has sophomore, junior, or senior standing.

If John has sophomore, junior, or senior standing, then he takes calculus,

Restate to Conditional Proposition

When you sing, my ears hurt.

Restate to Conditional Proposition

When you sing, my ears hurt.

If you sing, then my ears hurt.

Biconditional

Let p and q be two propositions. The proposition "**p** if and only if **q**" denoted by **$p \leftrightarrow q$** is called biconditional proposition.

$p \leftrightarrow q$ is true whenever the truth values of p and q are the same.

Biconditional : Truth Table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

“p if and only if q”

“p if q”

“p only if q”

"p if and only if q"

"p if q" = "if q then p"	"p only if q" = "if p then q"

"p if and only if q"

"p if q" = "if q then p"

$q \rightarrow p$

"p only if q" = "if p then q"

$p \rightarrow q$

"p if and only if q"

"p if q" = "if q then p"

$$q \rightarrow p$$

"p only if q" = "if p then q"

$$p \rightarrow q$$

$$(q \rightarrow p) \wedge (p \rightarrow q)$$

“p if and only if q”

“p if q” = “if q then p”

$q \to p$

“p only if q” = “if p then q”

$p \to q$

$$(q \to p) \land (p \to q) \equiv p \leftrightarrow q$$

Biconditional Representations

"p is necessary and sufficient for q and vice versa"

Biconditional Representations

"if p then q, and conversely"

Biconditional Representations

“p iff q”

Precedence of Logical Operators

Operator	*Precedence*
$\neg$	1
$\wedge$ $\vee$	2 3
$\rightarrow$ $\leftrightarrow$	4 5

Evaluate : p= T , q= F

$$p \rightarrow q \wedge \neg p$$

Evaluate : p= T , q= F

$$p \rightarrow q \land \neg p$$

Evaluate : p= T , q= F

$$p \rightarrow q \land \neg p$$

$$p \rightarrow q \land \neg T$$

Evaluate : p= T , q= F

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

Evaluate : p= T , q= F

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

Evaluate : p= T , q= F

$$p \rightarrow q \land \neg p$$

$$p \rightarrow q \land \neg T$$

$$p \rightarrow q \land F$$

$$p \rightarrow F \land F$$

Evaluate : p= T , q= F

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$

Evaluate : p= T , q= F

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$

Evaluate : p= T , q= F

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$

$$T \rightarrow F$$

Evaluate : p= T , q= F

$$p \rightarrow q \wedge \neg p$$

$$p \rightarrow q \wedge \neg T$$

$$p \rightarrow q \wedge F$$

$$p \rightarrow F \wedge F$$

$$p \rightarrow F$$

$$F \rightarrow F$$

$$F$$

Construct the truth table for the compound proposition

$$p \to q \land \neg p$$

Construct the truth table for the compound proposition

$$p \to q \land \neg p$$

p	q	$\neg p$	$q \land \neg p$	$p \to q \land \neg p$
T	T	F	F	F
T	F	F	F	F
F	T	T	T	T
F	F	T	F	T

Why do we translate English sentences into Logical Expressions?

Steps in translating English sentences into Logical Expressions

1. Find the connectives which are connecting propositions together.
2. Rename the propositions

Translate English sentences into Logical Expressions

- “You are not allowed to use a phone if your age is less than 18 years or you have no parent consent.”

Translate English sentences into Logical Expressions

Step 1:

- "You are not allowed to use a phone if your age is less than 18 years or you have no parent consent."

Translate English sentences into Logical Expressions

Step 2:

“You are not allowed to use a phone if your age is less than 18 years or you have no parent consent.”

Let:

q = “You are allowed to use a phone”
r =“Your age is less than 18 years”
s =“You have parent consent”

Translate English sentences into Logical Expressions

Step 2:

"You are not allowed to use a phone if your age is less than 18 years or you have no parent consent."

Let:

q = "You are allowed to use a phone"
r ="Your age is less than 18 years"
s ="You have parent consent"

$$(r \lor \neg s) \rightarrow \neg q$$

Are these statements consistent ?

“The system is in multiuser state if and only if it is operating normally.”

“If the system is operating normally, then the kernel is functioning.”

“The kernel is not functioning, or the system is in interrupt mode.”

“If the system is not in multiuser state, then it is in interrupt mode.”

“The system is not in interrupt mode.”

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

"If the system is operating normally, then the kernel is functioning."

"The kernel is not functioning, or the system is in interrupt mode."

"If the system is not in multiuser state, then it is in interrupt mode."

"The system is not in interrupt mode."

Let
p = "The system is in multiuser state."
q = "The system is operating normally."
r = "the kernel is functioning."
s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$$p \leftrightarrow q$$

"If the system is operating normally, then the kernel is functioning."

"The kernel is not functioning, or the system is in interrupt mode."

"If the system is not in multiuser state, then it is in interrupt mode."

"The system is not in interrupt mode."

Let
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Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$$p \leftrightarrow q$$

"If the system is operating normally, then the kernel is functioning."

$$q \rightarrow r$$

"The kernel is not functioning, or the system is in interrupt mode."

"If the system is not in multiuser state, then it is in interrupt mode."

"The system is not in interrupt mode."

Let

p = "The system is in multiuser state."

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s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$$p \leftrightarrow q$$

"If the system is operating normally, then the kernel is functioning."

$$q \rightarrow r$$

"The kernel is not functioning, or the system is in interrupt mode."

$$\neg r \vee s$$

"If the system is not in multiuser state, then it is in interrupt mode."

"The system is not in interrupt mode."

Let

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$$\neg p \rightarrow s$$

"The system is not in interrupt mode."

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"The kernel is not functioning, or the system is in interrupt mode."

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"If the system is not in multiuser state, then it is in interrupt mode."

$$\neg p \rightarrow s$$

"The system is not in interrupt mode."

$$\neg s$$

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Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$$p \leftrightarrow q$$

"If the system is operating normally, then the kernel is functioning."

$$q \rightarrow r$$

"The kernel is not functioning, or the system is in interrupt mode."

$$\neg r \vee s$$

"If the system is not in multiuser state, then it is in interrupt mode."

$$\neg p \rightarrow s$$

"The system is not in interrupt mode."

✓

$\neg s$ $\quad$ S = F

Let

p = "The system is in multiuser state."

q = "The system is operating normally."

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s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$p \leftrightarrow q$

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r$

"The kernel is not functioning, or the system is in interrupt mode."

$\neg r \vee s$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s$ F

"The system is not in interrupt mode." ✓

$\neg s$ S = F

Let

p = "The system is in multiuser state."

q = "The system is operating normally."

r = "The kernel is functioning."

s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$$p \leftrightarrow q$$

"If the system is operating normally, then the kernel is functioning."

$$q \rightarrow r$$

"The kernel is not functioning, or the system is in interrupt mode."

$$\neg r \lor s$$

"If the system is not in multiuser state, then it is in interrupt mode."

$$\neg p \rightarrow s^{F} \qquad P = T$$

"The system is not in interrupt mode."

✓

$$\neg s \qquad S = F$$

Let

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s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$p \leftrightarrow q$

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r$

"The kernel is not functioning, or the system is in interrupt mode."

$\neg r \vee s$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s^{F}$ $P = T$

"The system is not in interrupt mode."

$\neg s$ $S = F$

Let

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"The kernel is not functioning, or the system is in interrupt mode."

$\neg r \vee s^{F}$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s^{F}$ $P = T$

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$p \leftrightarrow q$

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r$

"The kernel is not functioning, or the system is in interrupt mode."

T $\neg r \vee s$ F

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s$ F P = T

"The system is not in interrupt mode."

$\neg s$ S = F

Let

p = "The system is in multiuser state."

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r = "The kernel is functioning."

s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$p \leftrightarrow q$

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r$

"The kernel is not functioning, or the system is in interrupt mode."

T $\neg r \vee s^{F}$ $r = F$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s^{F}$ $P = T$

"The system is not in interrupt mode."

$\neg s$ $S = F$

Let

p = "The system is in multiuser state."

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s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$p \leftrightarrow q$

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r$

"The kernel is not functioning, or the system is in interrupt mode."

$T\ \neg r \lor s^{F}$ $\quad r = F$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s^{F}$ $\quad P = T$

"The system is not in interrupt mode."

$\neg s$ $\quad S = F$

Let
p = "The system is in multiuser state."
q = "The system is operating normally."
r = "The kernel is functioning."
s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$p \leftrightarrow q$

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r^{F}$

"The kernel is not functioning, or the system is in interrupt mode."

T $\neg r \vee s^{F}$ $r = F$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s^{F}$ $P = T$

"The system is not in interrupt mode."

$\neg s$ $S = F$

Let

p = "The system is in multiuser state."

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r = "The kernel is functioning."

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Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$p \leftrightarrow q$

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r^{F}$ $\quad q = F$

"The kernel is not functioning, or the system is in interrupt mode."

$T\ \neg r \vee s^{F}$ $\quad r = F$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s^{F}$ $\quad P = T$

"The system is not in interrupt mode."

$\neg s$ $\quad S = F$

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"The kernel is not functioning, or the system is in interrupt mode."

$T\ \neg r \vee s^{F}$ $\quad r = F$

"If the system is not in multiuser state, then it is in interrupt mode."

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p = "The system is in multiuser state."
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Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

$p \leftrightarrow q$ F

"If the system is operating normally, then the kernel is functioning."

$q \rightarrow r$ F $q = F$

"The kernel is not functioning, or the system is in interrupt mode."

T $\neg r \vee s$ F $r = F$

"If the system is not in multiuser state, then it is in interrupt mode."

$\neg p \rightarrow s$ F $P = T$

"The system is not in interrupt mode."

$\neg s$ $S = F$

Let
p = "The system is in multiuser state."
q = "The system is operating normally."
r = "The kernel is functioning."
s = "The system is in interrupt mode."

Are these statements consistent ?

"The system is in multiuser state if and only if it is operating normally."

× $^{T}p \leftrightarrow q^{F}$

"If the system is operating normally, then the kernel is functioning."

✓ $q \rightarrow r^{F}$ $q = F$

"The kernel is not functioning, or the system is in interrupt mode."

✓ $^{T}\neg r \vee s^{F}$ $r = F$

"If the system is not in multiuser state, then it is in interrupt mode."

✓ $\neg p \rightarrow s^{F}$ $P = T$

"The system is not in interrupt mode."

✓ $\neg s$ $S = F$

Let

p = "The system is in multiuser state."

q = "The system is operating normally."

r = "The kernel is functioning."

s = "The system is in interrupt mode."

#1 .

Assume p and r = false, q and s = true

$$p \longrightarrow q$$

#2 .

Assume p and r = false, q and s = true

$$\neg p \rightarrow \neg q$$

#3 .

Assume p and r = false, q and s = true

$$\neg(p \rightarrow q)$$

#4.

Assume p and r = false, q and s = true

$$(p \to q) \wedge (q \to r)$$

#5.

Assume p and r = false, q and s = true

$$(p \rightarrow q) \rightarrow r$$

#6.

Assume p and r = false, q and s = true

$$p \rightarrow (q \rightarrow r)$$

#7.

Assume p and r = false, q and s = true

$$(s \rightarrow (p \wedge \neg r)) \wedge ((p \rightarrow (r \vee q)) \wedge s)$$

#8.

Assume p and r = false, q and s = true

$$((p \land \neg q) \to (q \land r)) \to (s \lor \neg q)$$

#9.

Assume p and r = false, q and s = true

$$((p \vee q) \wedge (q \vee s)) \rightarrow ((\neg r \vee p) \wedge (q \vee s))$$

#10.

Assume p and r = false, q and s = true

$$(q \vee r) \leftrightarrow r$$

Arguments

Assume p and r = false, q and s = true

$$(q \vee r) \leftrightarrow r$$

Implication vs Converse vs Contrapositive vs Inverse

Implication: $p \rightarrow q$

Converse: $q \rightarrow p$

Contrapositive: $\neg q \rightarrow \neg p$

Inverse: $\neg p \rightarrow \neg q$

Implication vs Converse vs Contrapositive vs Inverse

- Implication: $p \rightarrow q$
  "If it rains today then, I will stay at home."

Implication vs Converse vs Contrapositive vs Inverse

Implication: $p \rightarrow q$

"If it rains today then, I will stay at home."

Converse: $q \rightarrow p$

Implication vs Converse vs Contrapositive vs Inverse

Implication: **p → q**

"If it rains today then, I will stay at home."

Converse: **q → p**

"If I will stay at home, then it rains today."

Implication vs Converse vs Contrapositive vs Inverse

Implication: **p → q**

"If it rains today then, I will stay at home."

Converse: **q → p**

"If I will stay at home, then it rains today."

Implication: **$p \rightarrow q$**

"If it rains today then, I will stay at home."

Converse: **$q \rightarrow p$**

"If I will stay at home, then it rains today."

Contrapositive: **$\neg q \rightarrow \neg p$**

Implication: $p \rightarrow q$

"If it rains today then, I will stay at home."

Converse: $q \rightarrow p$

"If I will stay at home, then it rains today."

Contrapositive: $\neg q \rightarrow \neg p$

"If I will not stay at home, then it does not rain today."

Implication: $p \rightarrow q$

"If it rains today then, I will stay at home."

Converse: $q \rightarrow p$

"If I will stay at home, then it rains today."

Contrapositive: $\neg q \rightarrow \neg p$

"If I will not stay at home, then it does not rain today."

Inverse: $\neg p \rightarrow \neg q$

Implication: **p → q**

“If it rains today then, I will stay at home.”

Converse: **q → p**

“If I will stay at home, then it rains today.”

Contrapositive: **¬q → ¬p**

“If I will not stay at home, then it does not rain today.”

Inverse: **¬p → ¬q**

“If it does not rain today then, I will not stay at home.”

Implication($p \rightarrow q$) and Contrapositive($\neg q \rightarrow \neg p$) are equivalent.

Implication: “If it rains today then, I will stay at home.”
Contrapositive: “If I will not stay at home, then it does not rain today.”

Converse ($q \rightarrow p$) and Inverse($\neg p \rightarrow \neg q$) are equivalent.

Converse: “If I will stay at home, then it rains today.”
Inverse:“If it does not rain today then, I will not stay at home.”

- Neither Converse (**q → p**) nor Inverse(**¬p → ¬q**) is equivalent to Implication (**p → q**).

Implication: ”If it rains today then, I will stay at home.”
Converse: “If I will stay at home, then it rains today.”
Inverse:“If it does not rain today then, I will not stay at home.”

p	q	$p \to q$ (Implication)	$q \to p$ (converse)	$\sim p \to \sim q$ (inverse)	$\sim q \to \sim p$ (contrapositive)
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Implication(p → q) and Contrapositive(¬q → ¬p) are equivalent.

p	q	$p \rightarrow q$ (Implication)	$q \rightarrow p$ (converse)	$\sim p \rightarrow \sim q$ (inverse)	$\sim q \rightarrow \sim p$ (contrapositive)
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Converse (q → p) and Inverse(¬p → ¬q) are equivalent.

p	q	$p \to q$ (Implication)	$q \to p$ (converse)	$\sim p \to \sim q$ (inverse)	$\sim q \to \sim p$ (contrapositive)
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Neither Converse (q → p) nor Inverse(¬p → ¬q) is equivalent to Implication (p → q).

Tautology

Compound proposition which is **always true**.

P	~P		P v ~P
T	F		T
F	T		T

Contradiction

Compound proposition which is **always false**.

P	~P	P ∧ ~P
T	F	F
F	T	F

Contingency

Compound proposition which is **sometimes true** and **sometimes false**.

x	y	x ∧ y
T	T	T
T	F	F
F	T	F
F	F	F

Satisfiable

Compound proposition which has at least **one true** result.

x	y	x ∧ y
T	T	T
T	F	F
F	T	F
F	F	F

True or False ?

Tautology is always satisfiable.

True or False ?

Tautology is always satisfiable and satisfiable is always tautology.

True or False ?

Contradiction is always unsatisfiable.

Valid

Compound proposition is valid when it is a tautology.

P	~P	P v ~P
T	F	T
F	T	T

Invalid

Compound proposition which is either a contradiction or contingency.

P	~P	P ∧ ~P
T	F	F
F	T	F

Logical Equivalences

Two different propositions are equivalent if they have the same truth values.

• Logical Equivalences

Two different propositions are equivalent if they have the same truth values.

p	q	$\neg(p \to q)$	$p \wedge \neg q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	F	F

Logical Equivalences

Two different propositions are equivalent if they have the same truth values.

p	q	$\neg(p \to q)$	$p \land \neg q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	F	F

Logical Equivalences

$$p \equiv q$$

Table 4-1 Laws of the algebra of propositions

Idempotent laws:	(1a) $p \vee p \equiv p$	(1b) $p \wedge p \equiv p$
Associative laws:	(2a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$	(2b) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Commutative laws:	(3a) $p \vee q \equiv q \vee p$	(3b) $p \wedge q \equiv q \wedge p$
Distributive laws:	(4a) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	(4b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
Identity laws:	(5a) $p \vee F \equiv p$ (6a) $p \vee T \equiv T$	(5b) $p \wedge T \equiv p$ (6b) $p \wedge F \equiv F$
Involution law:	(7) $\neg\neg p \equiv p$	
Complement laws:	(8a) $p \vee \neg p \equiv T$ (9a) $\neg T \equiv F$	(8b) $p \wedge \neg p \equiv T$ (9b) $\neg F \equiv T$
DeMorgan's laws:	(10a) $\neg(p \vee q) \equiv \neg p \wedge \neg q$	(10b) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

Rules of Inference

Premise

Basis for the conclusion / evidence /assumption

Conclusion

Reached from the given set of premises.

If premise then conclusion

Argument

Set of one or more premises and a conclusion.

Valid Argument

If premises are all true, then conclusion must also be true.

Is it a valid argument ?

1. All humans are mortal. (Premise)
2. Socrates is a human. (Premise)
3. Therefore, Socrates is mortal. (Conclusion)

Is it a valid argument ?

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

Translate into Logical Expressions

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p \rightarrow q$

Translate into Logical Expressions

“If I love cat, then I love dog.”
“I love cat.”
Therefore, “I love dog.”

$p \rightarrow q$

p

Translate into Logical Expressions

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$$\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Translate into Logical Expressions

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p \rightarrow q$
p

$\therefore q$

$$((p \rightarrow q) \land p) \rightarrow q$$

Identify if it is a Valid Argument

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p \to q$ T

p T

$\therefore q$

Assume all premises are true, we need to make conclusion true.

Identify if it is a Valid Argument

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p \rightarrow q$	T
p T	T
$\therefore q$	T

Identify if it is a Valid Argument

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p^{T} \rightarrow q$	T
p^{T}	T
$\therefore q$	T

Identify if it is a Valid Argument

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p^{T} \rightarrow q$	T
p^{T}	T
$\therefore q$	T

What is the value of q, that will make the first premise true ?

Identify if it is a Valid Argument

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p^{T} \rightarrow q^{T}$	T
p^{T}	T
$\therefore q$	T

What is the value of q, that will make the first premise true ?

Identify if it is a Valid Argument

"If I love cat, then I love dog."
"I love cat."
Therefore, "I love dog."

$p^{T} \rightarrow q^{T}$	T
p^{T}	T
$\therefore q^{T}$	T

If all premises and conclusion are true, then it is a valid argument.

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p \rightarrow q$

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p \rightarrow q$
q

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$$\begin{array}{l} p \rightarrow q \\ q \\ \hline \therefore p \end{array}$$

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p \rightarrow q$ T

q T

$\therefore p$

Assume all premises are true, we need to make conclusion true.

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p \rightarrow q$ T

q T T

$\therefore p$

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p \rightarrow q^{T}$	T
q^{T}	T

$\therefore p$

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p \rightarrow q^{T}$ T

q^{T} T

$\therefore p$

What is the value of p that will make the first premise true ?

Is it a valid argument ?

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p \rightarrow q^{T}$ T

q^{T} T

$\therefore p$

What is the value of p that will make the first premise true ? **T / F**

Is it a valid argument ?

If p is true

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p^{T} \rightarrow q^{T}$	T
q^{T}	T

$\therefore p$

What is the value of p that will make the first premise true ? **T / F**

Is it a valid argument ?

If p is true

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p^{T} \rightarrow q^{T}$ T
q^{T} T
____
$\therefore p^{T}$

What is the value of p that will make the first premise true ? **T / F**

Is it a valid argument ?

If p is false

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p^{F} \rightarrow q^{T}$ T

q^{T} T

$\therefore p^{F}$

What is the value of p that will make the first premise true ? **T / F**

Is it a valid argument ?

If p is false, then all premises are true and the conclusion is false

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p^{F} \rightarrow q^{T}$ T

q^{T} T

$\therefore p^{F}$

What is the value of p that will make the first premise true ? **T / F**

Is it a valid argument ?

If p is false, then all premises are true and the conclusion is false

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p^{F} \rightarrow q^{T}$ T

q^{T} T

$\therefore p^{F}$

What is the value of p that will make the first premise true ? **T / F**

Is it a valid argument ?

If p is false, then all premises are true and the conclusion is false

"If I love cat, then I love dog."
"I love dog."
Therefore, "I love cat."

$p^{F} \rightarrow q^{T}$ T

q^{T} T

$\therefore p^{F}$

What is the value of p that will make the first premise true ? **T / F**

Inference

Deriving conclusions from evidences

Rules of Inference

Templates for constructing valid arguments.

Types of Inferences

1. Modus Ponens
2. Modus Tollens
3. Hypothetical Syllogism
4. Disjunctive Syllogism
5. Addition
6. Simplification
7. Conjunction
8. Resolution

Modus Ponens

$((p \rightarrow q) \land p) \rightarrow q$

$$\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Modus Ponens

$((p \rightarrow q) \wedge p) \rightarrow q$

$p \rightarrow q$	T
p	T

$\therefore q$

Modus Ponens

$((p \to q) \land p) \to q$

$p \to q$	T
p T	T

$\therefore q$

Modus Ponens

$((p \rightarrow q) \wedge p) \rightarrow q$

$p^{T} \rightarrow q$	T
p^{T}	T

$\therefore q$

What is the value of q that will make the first premise true ?

Modus Ponens

$((p \to q) \land p) \to q$

$p^{T} \to q$	T
p T	T

$\therefore q$

What is the value of q that will make the first premise true ? T

Modus Ponens

$$((p \rightarrow q) \land p) \rightarrow q$$

$\boxed{p^{T} \rightarrow q^{T}}$ T

$\boxed{p^{T}}$ T

$\therefore q$

Modus Ponens

$((p \to q) \land p) \to q$

$\boxed{p^{T} \to q^{T}}$ T

$\boxed{p^{T}}$ T

$\therefore q^{T}$

We have proven that conclusion is true given all premises are true.

Modus Tollens

$((p \rightarrow q) \land \neg q) \rightarrow \neg p$

$p \rightarrow q$	T
$\neg q$	T

$\therefore \neg p$

Hypothetical Syllogism

$$[(p \to q) \wedge (q \to r)] \to (p \to r)$$

$$\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore p \to r \end{array}$$

Disjunctive Syllogism

$((p \lor q) \land \neg p) \to q$

$$
\begin{array}{l}
p \lor q \\
\neg p \\
\hline
\therefore q
\end{array}
$$

Addition

$(p) \rightarrow p \vee q$

$$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$$

Simplification

$(p \land q) \to p$

$$\begin{array}{l} p \land q \\ \hline \therefore p \end{array} \quad \text{or} \quad \begin{array}{l} p \land q \\ \hline \therefore q \end{array}$$

Conjunction

$[(p) \land (q)] \rightarrow (p \land q)$

$$
\begin{array}{l}
p \\
q \\
\hline
\therefore p \land q
\end{array}
$$

Resolution

$(p \lor q) \land (\neg p \lor r) \rightarrow (q \lor r)$

$p \lor q$

$\neg p \lor r$

$\therefore q \lor r$

Thanks!

Any questions?

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References

Schaum's Outlines, Discrete Mathematics (Third Edition)

Credits

Special thanks to all the people who made and released these awesome resources for free:

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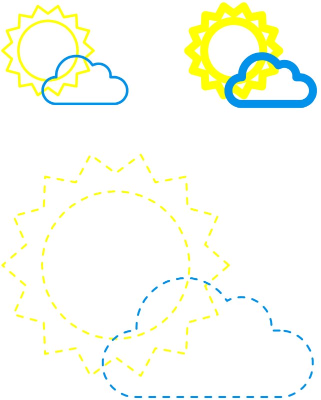

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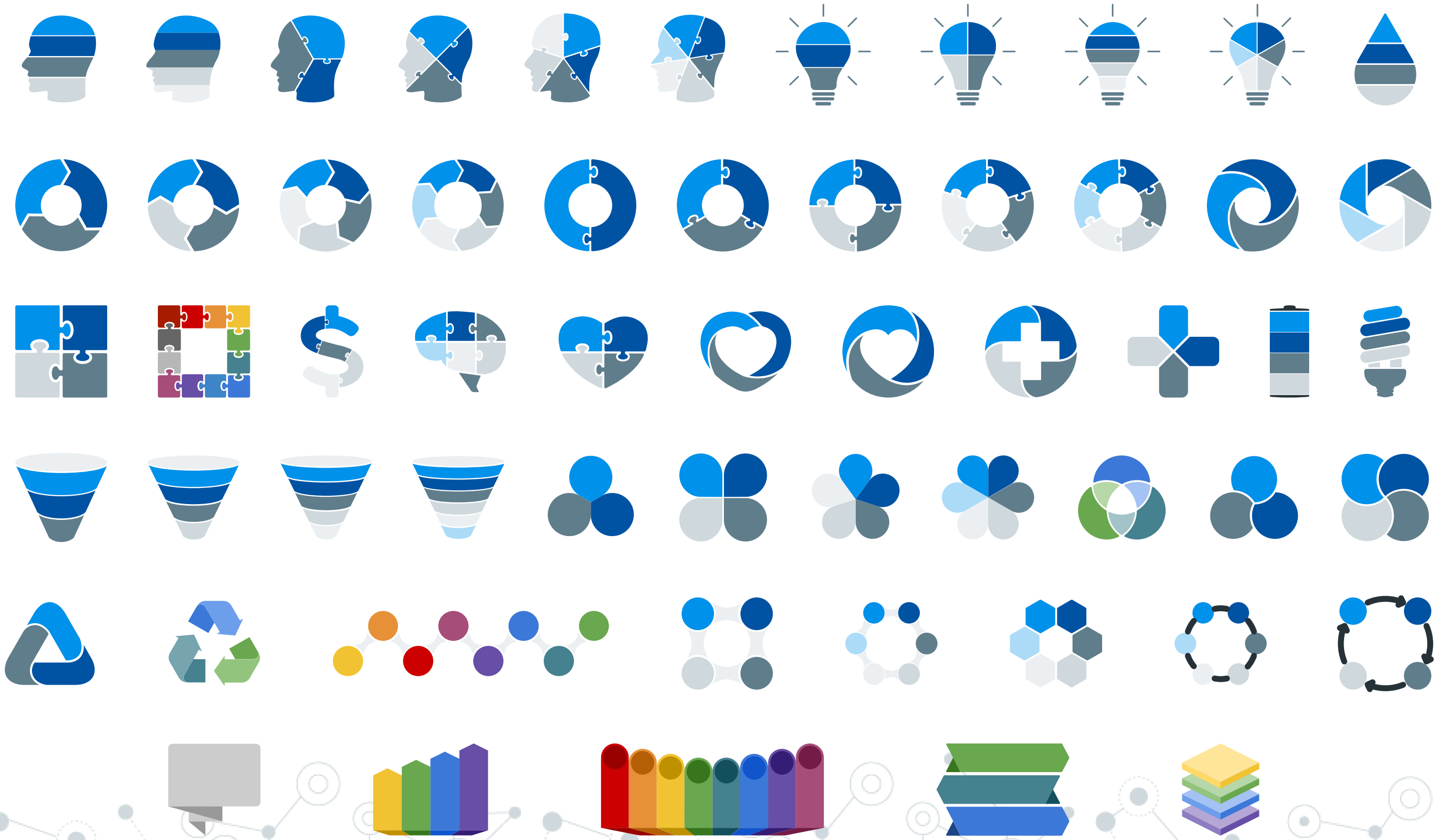

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